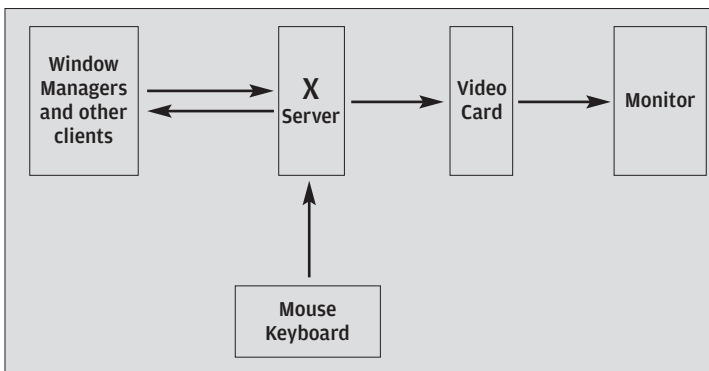


4.1 The X Window System

We will now look at what the X Window System is, and give a background of how the GUI came into Linux.



The X Server–Client Model

If you have observed different applications in Windows, you will find that all applications behave the same with respect to their window positions and the options they give the user to change the way they look (maximise, minimise, resize, drag-and-drop, etc.). This is because all applications use the same set of display functions that are built into Windows. For example, the developers simply need to “call” a particular inbuilt function in Windows for the Maximise, Minimise and Close buttons. All programs would use the same function—hence the uniformity. (If a program has to look different, the developers will have to write display functions of their own.) In Linux, however, you will see that different applications look and feel quite different from each other. Each “window” or application can have its own features. The way in which a user can modify the appearance of the window can also vary. This is due to the versatility that the X Window system offers to the different desktop environments, and the different tool kits that are available for developers. For example, if a developer so wishes to, he can write his program such that double-clicking on the title bar rolls